

Experiment ‘p_2_2_letter_perimeter’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_2_letter_perimeter

Natural language statement: null

Method signature: p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)

Ensures

- `perimeter == (8.5 + 11.0) * 2.0`
- `-0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_2_letter_perimeter()` returns (perimeter: real, diagonal: real)

The method should respect the following contract:

ensures perimeter == (8.5 + 11.0) * 2.0, ensures -0.01 < (diagonal * diagonal) - ((8.5 *
↪ 8.5) + (11.0 * 11.0)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2fsE0DOSs1w7avoLDyLEryxliHK

Model: gpt-5.5-2026-04-23

Created at: 1780580724

Finish reason: stop

Usage: 354 tokens in, and 665 tokens out

```
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9015;
  assert diagonal * diagonal == 193.25170225;
  assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;
  assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == 0.00170225;
  assert -0.01 < 0.00170225 < 0.01;
}
```

Final Program

```
method p_2_2_letter_perimeter() returns (perimeter: real, diagonal: real)
  ensures perimeter == (8.5 + 11.0) * 2.0
  ensures -0.01 < (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) < 0.01
{
  perimeter := (8.5 + 11.0) * 2.0;
  diagonal := 13.9015;
  assert diagonal * diagonal == 193.25170225;
```

```
assert (8.5 * 8.5) + (11.0 * 11.0) == 193.25;  
assert (diagonal * diagonal) - ((8.5 * 8.5) + (11.0 * 11.0)) == 0.00170225;  
assert -0.01 < 0.00170225 < 0.01;  
}
```

Total Token Usage

Input tokens: 354

Output tokens: 665

Reasoning tokens: 512

Sum of 'total tokens': 1019

Experiment Timings

Overall Experiment started at 1780580722636, ended at 1780580771285, lasting 48649ms (48.65 seconds)

Iteration #1 started at 1780580722685, ended at 1780580771261, lasting 48576ms (48.58 seconds)